

**17**

$$v = r\omega$$

$$\omega = \frac{v}{r} = \frac{1.0}{0.5} = 2 \text{ rad s}^{-1}$$

$$\begin{aligned}x &= x_0 \cos(\omega t) \\&= (0.50) \cos(2(0.9)) \\&= -0.11 \text{ m}\end{aligned}$$

$$\text{distance travelled} = x_0 - (-0.11) = 0.61 \text{ m}$$

**18** no constant phase difference